

Chemistry

Units 3 and 4

Practice Exam Answers

Section 1: Multiple choice

(25 marks)

Question 1

- B** Statement II is incorrect. The energy of collision must be equal to or greater than the activation energy.

Question 2

- C** The equilibrium law expression for the equation $aA + bB \rightarrow cC + dD$ is $K_c = \frac{[C]^c[D]^d}{[A]^a[B]^b}$.

Question 3

- C** The value of the equilibrium constant of the reverse reaction is the reciprocal of the value for the forward reaction (i.e. $\frac{1}{25}$ or 0.04).

Question 4

- B** The value of the equilibrium constant is greater at the higher temperature, indicating that the reaction is endothermic and the activation energy of the forward reaction is greater than that for the reverse reaction. Higher temperatures favour the forward reaction, resulting in an increase in the yield of N_2O . In any system, at equilibrium the rate of the forward and reverse reactions are equal.

Question 5

- D** The low value of the equilibrium constant indicates that the concentration of the products, N_2O , will be less than the concentration of the reactants.

Question 6

- D** In the reaction, NH_4^+ acts as an acid because it donates a proton to the S^{2-} ion. The conjugate base, NH_3 , has one proton less than the acid, NH_4^+ .

Question 7

- A** The definition of pH is $pH = -\log_{10}[H_3O^+]$. Since HCl is a strong acid and completely ionised in solution, it can be assumed that in each beaker $[HCl] = [H_3O^+]$. The pH is only dependent on the $[H_3O^+]$, not the volume of solution. Because the definition involves the negative logarithm, the pH increases as the $[H_3O^+]$ decreases. The solution that has the lowest $[H_3O^+]$ will have the highest pH.

Question 8

- A** The pH is dependent on the concentration of hydronium ions. HCl in beaker A is a strong acid and is completely ionised. The addition of 10 mL of water will double the volume and consequently halve the concentration of H_3O^+ .

The CH_3COOH in beaker B is a weak acid and is only partly ionised. The concentration of H_3O^+ in beaker B is less than the concentration of H_3O^+ in beaker A and consequently the pH change will be less when this solution is diluted.

The volume of solution in beaker C is greater than that in beakers A and B. The addition of 10 mL of water will dilute this solution to a lesser extent than the addition of 10 mL of water to the other two beakers.

Question 9

- C** Pure water, at any temperature, is neutral because the concentrations of hydronium ions and hydroxide ions are equal. At 1°C , $[\text{H}_3\text{O}^+] = [\text{OH}^-] = 1.0 \times 10^{-7.5} \text{ mol L}^{-1}$. The equilibrium constant for the self-ionisation of water is as follows.

$$K_w = [\text{H}_3\text{O}^+][\text{OH}^-] = 1.0 \times 10^{-7.5} \times 1.0 \times 10^{-7.5} = 10^{-15}$$

Question 10

- D** The indicator should change colour at the equivalence point in the titration when the amount, in moles, of $\text{H}_3\text{O}^+ = \text{OH}^-$.

Question 11

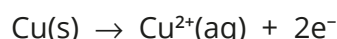
- B** The burette scale is read from the top down. The volume is read from the bottom of the meniscus and with practice can be judged to the nearest 0.05 mL.

Question 12

- A** The oxidation number of both Cr in Cr_2O_3 and AlCl_3 is +3. The oxidation number of N in NH_3 is -3 and that of S in SO_3^{2-} is +4. Mn in MnO_4^- has an oxidation number of +7.

Question 13

- D** The mass of copper electrode decreases due to the oxidation of copper as per the following equation. The copper electrode has a negative charge as electrons are produced.



Question 14

- C** From their position in the Table of Standard Reduction Potential, at 25°C it is evident that Cl_2 is the stronger oxidant and Fe^{2+} the stronger reductant.

The potential difference produced is given by the following expression.

$$E^\circ_{\text{cell}} = E^\circ_{\text{oxidant}} - E^\circ_{\text{reductant}} = 1.36 - 0.77 = 0.59 \text{ V}$$

The value of the cell potential is not dependent on the coefficients in the overall cell equation.

Question 15

- D** A spontaneous reaction occurs between the oxidising agent that has a higher E° and the reducing agent that has the lower E° . The oxidising agent, Fe^{3+} in the $\text{Fe}^{3+}/\text{Fe}^{2+}$ redox pair, has a higher E° (0.77 V) than the reducing agent, Zn, in the Zn^{2+}/Zn redox pair (-0.76 V).

Question 16

- B** Impurities such as silver and gold that are less reactive (weaker reducing agents) than copper do not undergo a reaction at the anode but fall off and are deposited underneath the anode. Copper and impurities that are more reactive than copper are oxidised at the anode.

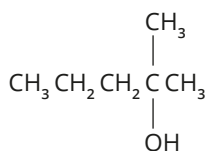
Question 17

- B** The functional groups are hydroxyl (OH), carboxyl (COOH) and amine (NH₂).

Question 18

- C** The compound has an alcohol (OH), functional group bonded to the second atom on a 5-carbon atom chain. The stem is pentan-2-ol. Numbering starts from the end that results in the use of the smallest possible number to indicate the position of the OH group, so pentan-4-ol is not correct.

There is also a methyl (CH₃) side chain attached to the 5-carbon chain. This is indicated as a prefix attached to the name of the main chain, methylpentan-2-ol. The position of the methyl group is indicated by the number of the carbon atom to which it is attached. The full name of the compound is 2-methylpentan-2-ol.



Question 19

- D** An alkene such as propene undergoes an addition reaction with chlorine. In this reaction, two chlorine atoms are added to one of the bonds in the double bond. A dichloroalkane, in this case 1,2-dichloropropane, is formed.

Question 20

- D** The compound is an ester formed between the reaction of propan-1-ol (CH₃CH₂CH₂OH) and ethanoic (acetic) acid (CH₃COOH).

Question 21

- C** The polymer is an addition polymer formed by an addition reaction between alternating CH₂CHCl and CH₂CCl₂ monomers.

Question 22

- D** At pH 2 the basic amine (NH₂) groups have accepted a proton to form NH₃⁺ groups. The carboxylic acid (COOH) group has not ionised and remains as COOH.

Question 23

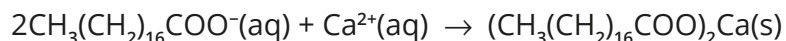
- C** The catalysed reaction occurs on the surface of the cobalt catalyst. The powdered form of the catalyst provides a greater surface area on which the reaction can occur, resulting in a greater rate of reaction.

Question 24

- A** The value of the equilibrium constant for an exothermic reaction decreases as the temperature increases (i.e. the yield of product is greater at lower temperatures). An excess of chlorine will shift the position of equilibrium to the right, increasing the amount of product. High pressures will cause the equilibrium to favour the direction that produces the least number of molecules (i.e. the forward reaction).

Question 25

- B** Hard water contains Ca^{2+} , Mg^{2+} or Fe^{2+} ions. Each of these ions can form a precipitate with the fatty acid anions found in soaps:



The triglyceride in alternative A and the dodecacylbenzene in alternative C are insoluble in both hard and soft water. Sodium alkylbenzene sulfonate, shown in alternative D, is used in detergents and does not form a precipitate with Ca^{2+} , Mg^{2+} or Fe^{2+} ions.

End of section 1

Section 2: Short answer

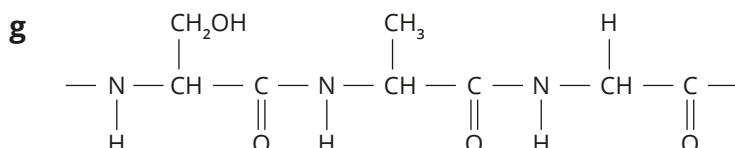
(74 marks)

(Note: In multi-mark questions each allocated mark is indicated as * and total marks are indicated in parentheses.)

Question 26

- a** Addition polymerisation occurs between monomers that have a double bond between two carbon atoms (i.e. monomers II, IV, VI and IX). (1 mark)
- b** monomer IV, tetrafluorethene polymerised to form poly(tetrafluorethene) or Teflon (1 mark)
- c** monomers III, V and XI, because they are all amino acids (1 mark)
- d** monomers II and VI (1 mark)
- e** polyamide (1 mark)
- f i** monomer XII (1 mark)
- ii** condensation polymerisation*

Monomer XII has both the OH functional group and the COOH functional groups. It undergoes a condensation reaction when the OH group on one monomer reacts with the COOH group of an adjacent monomer forming an ester (COO) bond*. (2 marks)



The abbreviations for amino acids as shown in the Data Book can be used to represent this sequence of amino acids as ser-ala-gly.

The combination of the three amino acid residues in any order is acceptable*.

The peptide of amide bond (CONH) must be correctly shown connecting the residues*. (2 marks)

Question 27

- a**
- | Primary alcohol | Secondary alcohol |
|--|--|
| $\begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \\ \quad \quad \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{OH} \\ \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \end{array}$ | $\begin{array}{c} \text{H} \quad \text{OH} \quad \text{H} \\ \quad \quad \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{H} \\ \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \end{array}$ |
- (2 marks)
- b**
- | Name | Structural formula |
|----------|--|
| propanal | $\begin{array}{c} \text{H} \quad \text{H} \quad \text{O} \\ \quad \quad // \\ \text{H}-\text{C}-\text{C}-\text{C} \\ \quad \quad \backslash \\ \text{H} \quad \text{H} \quad \text{H} \end{array}$ |
| Category | |
| aldehyde | |
- (3 marks)
- c**
- | Name | Structural formula |
|-----------|---|
| propanone | $\begin{array}{c} \text{H} \quad \text{O} \quad \text{H} \\ \quad \quad \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{H} \\ \quad \quad \\ \text{H} \quad \quad \text{H} \end{array}$ |
| Category | |
| ketone | |
- (3 marks)

d i orange to green (1 mark)

ii $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH(l)} + \text{H}_2\text{O(l)} \rightarrow \text{CH}_3\text{CH}_2\text{COOH(l)} + 4\text{H}^+(\text{aq}) + 4\text{e}^-$ (1 mark)

Question 28

a ethyl butanoate (1 mark)

b i $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH} + \text{CH}_3\text{CH}_2\text{OH} \rightleftharpoons \text{CH}_3\text{CH}_2\text{CH}_2\text{COOCH}_2\text{CH}_3 + \text{H}_2\text{O}$

$$n(\text{butanoic acid}) = \frac{0.8811}{88.11} = 0.010\,00\text{ mol}$$

$$n(\text{ethanol}) = \frac{0.7410}{46.07} = 0.016\,08^*$$

Assuming that you had 0.01 mol of butanoic acid, you would need 0.01 mol of ethanol. You have 0.0160 mol of ethanol, therefore:

Ethanol is in excess and butanoic acid is the limiting reagent*. (2 marks)

ii Theoretical mass of flavouring agent produced is:

$$\begin{aligned} n(\text{flavouring agent}) &= n(\text{butanoic acid}) \\ &= 0.010\,00^* \end{aligned}$$

$$\begin{aligned} m(\text{flavouring agent}) &= 0.010\,00 \times 116.2 \\ &= 1.162\text{ g}^* \end{aligned}$$

$$\text{actual mass} = 0.907\text{ g}$$

$$\begin{aligned} \% \text{ yield} &= \frac{0.907}{1.162} \times 100 \\ &= 78.1\%^* \end{aligned}$$

(3 marks)

Question 29

a i $\text{NH}_4^+(\text{aq}) + \text{H}_2\text{O(l)} \rightleftharpoons \text{NH}_3(\text{aq}) + \text{H}_3\text{O}^+(\text{aq})$ (1 mark)

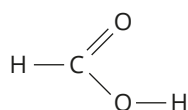
ii $\text{CO}_3^{2-}(\text{aq}) + \text{H}_2\text{O(l)} \rightleftharpoons \text{HCO}_3^-(\text{aq}) + \text{OH}^-(\text{aq})$ (1 mark)

iii $\text{CO}_3^{2-}/\text{HCO}_3^-$ or $\text{H}_2\text{O}/\text{OH}^-$ * (1 mark)

b Both reactions occur when ammonium carbonate dissolves in water. In reaction ii, the hydrolysis of carbonate ion occurs to a greater extent. More OH^- ions than H_3O^+ ions are produced and the solution has a pH greater than 7. (1 mark)

Question 30

a methanoic acid



(1 mark)

b An acid is a proton donor. The proton that is donated by an acid is attached to an electronegative element such as oxygen. The bond between the oxygen and the proton is polar. This bond is broken when methanoic acid undergoes an ionisation reaction with water*.

The bond between C and H is non-polar and the proton is not donated to a water molecule so methanoic acid is monoprotic*. (2 marks)

Question 31

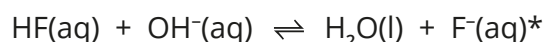
- a $\text{pH} = -\log[0.001\ 34] = 2.87$ (1 mark)
- b i beaker B with hypochlorous acid (1 mark)
- ii beaker A with ethanoic acid (1 mark)
- iii beaker A due to greater ionisation (1 mark)

Question 32

- a $\text{HF(aq)} + \text{H}_2\text{O(l)} \rightleftharpoons \text{H}_3\text{O}^+(\text{aq}) + \text{F}^-(\text{aq})$ (1 mark)
- b $K_a = \frac{[\text{H}_3\text{O}^+][\text{F}^-]}{[\text{HF}]}$ (1 mark)
- c From the acidity constant expression:

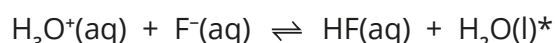
$$[\text{H}_3\text{O}^+] = K_a \times \frac{[\text{HF}]}{[\text{F}^-]}$$

The buffer contains a large amount of the acid HF and the base F^- . The concentration of H_3O^+ of this buffer, and hence its pH, is determined by the ratio $[\text{HF}]/[\text{F}^-]$. The OH^- ions, from the addition of a strong base, will react with the weak acid (HF):



The addition of a small amount of OH^- causes a small reduction in the concentration of HF and a small increase in the concentration of F^- . This causes only a slight change in the ratio $[\text{HF}]/[\text{F}^-]$ and consequently a slight change in pH^* .

H_3O^+ ions, from the addition of a small amount of strong acid, will react with F^- ions:



The concentration of F^- decreases slightly and the concentration of HF increases slightly. The change in $[\text{HF}]/[\text{F}^-]$ is small and there is only a slight change in pH^* .

The change in pH is small as long as the buffer contains relatively large amounts of HF and F^- in comparison to any added H_3O^+ or OH^- . (4 marks)

Question 33

- a $\text{H}_2\text{O}_2(\text{l}) + 2\text{H}^+(\text{aq}) + 2\text{e}^- \rightleftharpoons 2\text{H}_2\text{O(l)}$
 $\text{H}_2\text{O}_2(\text{l}) \rightleftharpoons \text{O}_2(\text{g}) + 2\text{H}^+(\text{aq}) + 2\text{e}^-$

The overall equation is:



- b Even though there is a large difference in the standard reduction potentials for the oxidation and reduction of H_2O_2 , the rate of decomposition is slow.

Spontaneous reactions such as decomposition of H_2O_2 occur without intervention. However, spontaneous reactions do not always occur instantaneously (e.g. rusting of iron or decomposition of H_2O_2). Standard electrode potentials do not give an indication of rate of reaction.

Rate of decomposition of H_2O_2 can be increased by exposure to light. Storage in a dark container limits exposure to light. (1 mark)

H₂O₂ is ...	by reactant	Equation
oxidised	the oxidant of any redox pair that has E° greater than 0.70 V (e.g. Cl ₂)	$\text{H}_2\text{O}_2(\text{l}) + \text{Cl}_2(\text{g}) \rightleftharpoons \text{O}_2(\text{g}) + 2\text{H}^+(\text{aq}) + 2\text{Cl}^-(\text{aq})$ (1 mark for any appropriate balanced equation)
reduced	the reductant of any redox conjugate pair that has E° below 1.76 V (e.g. I ⁻ ions)	$\text{H}_2\text{O}_2(\text{l}) + 2\text{H}^+(\text{aq}) + 2\text{I}^-(\text{aq}) \rightleftharpoons 2\text{H}_2\text{O}(\text{l}) + \text{I}_2(\text{s})$ (1 mark for any appropriate balanced equation)

(4 marks)

Question 34

- a i** Highest voltage can be produced by combining the half-cell with the highest E° with the half-cell that has the lowest E° value*.

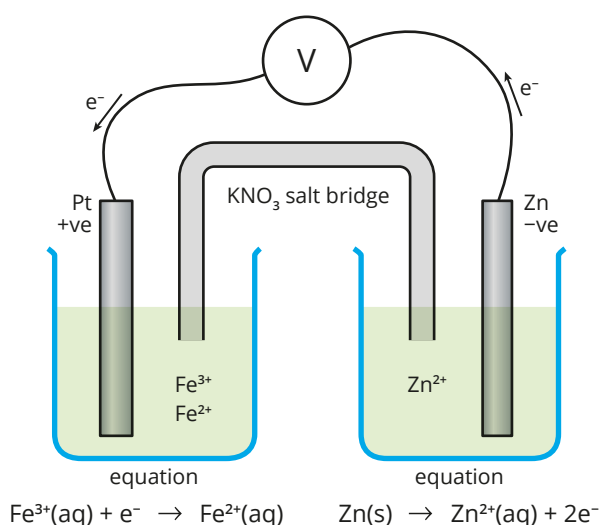
In this case, the Fe³⁺/Fe²⁺ and Zn/Zn²⁺ half-cells*.

(2 marks)

ii $0.77 - (-0.76) = 1.53 \text{ V}$

(1 mark)

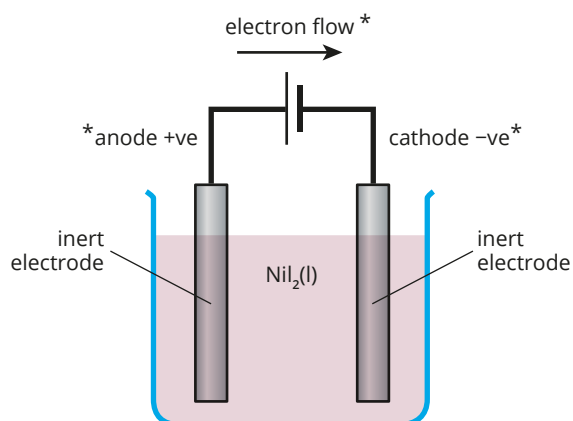
b



(5 marks, 1 mark each for correct direction of electron flow, correct polarity of electrodes, correct reduction equation, correct oxidation equation and inclusion of salt bridge to complete the circuit)

Question 35

a and b



(3 marks)

- c i** anode reaction: $2\text{I}^-(\text{l}) \rightarrow \text{I}_2(\text{l}) + 2\text{e}^-$

(1 mark)



(1 mark)

- ii** overall reaction: $2\text{I}^-(\text{l}) + \text{Ni}^{2+}(\text{l}) \rightarrow \text{I}_2(\text{l}) + \text{Ni}(\text{l})$

(1 mark)

Question 36

Structure	Bonding and interactions
primary	The primary structure of a protein is the sequence of amino acids in the polypeptide chain. This is maintained by covalent bonding. The amino acid residues are connected by amide or peptide functional groups (CONH). These are formed by a condensation reaction between the carboxylic acid functional group (COOH) in one amino acid molecule and the amine functional group (NH ₂) in an adjacent amino acid*.
secondary	There are two main types of secondary structures. An α-helix is formed by hydrogen bonding between the H in an amide functional group and the oxygen of the carbonyl group (C=O) in another, nearby, amide group*. The β-pleated sheet structure is a result of H bonding between the N—H and C=O groups on an adjacent polypeptide chain*.
tertiary	The tertiary structure refers to the unique folding of the protein molecule. Tertiary structure is maintained by interaction between side chains on amino acid residues*. The nature of interaction depends on the nature of the side chain and includes covalent H-bonding, dipole–dipole interaction, dispersion forces, ionic interactions and disulphide bridges*.

(5 marks)

Question 37

- a i** $\text{CH}_3\text{CH}_2(\text{g}) + \text{H}_2\text{O}(\text{g}) \xrightarrow{300^\circ\text{C} / \text{high pressure } \text{H}_3\text{PO}_4 \text{ catalyst}} \text{CH}_3\text{CH}_2\text{OH}(\text{g})$ (1 mark)
- ii** $\text{C}_6\text{H}_{12}\text{O}_6(\text{aq}) \xrightarrow{\text{enzymes}} 2\text{CH}_3\text{CH}_2\text{OH}(\text{aq}) + 2\text{CO}_2(\text{g})$ (1 mark)
- b** The reaction pathway of a catalysed reaction has a lower activation energy*. More particles will have sufficient energy to react, thereby increasing the rate*. (2 marks)
- c i** Lower temperatures and pressures are required in enzyme-catalysed reactions and costly high temperature/pressure equipment is not required. (1 mark)
- ii** Ethene is derived from cracking of alkanes, which are found in hydrocarbon deposits. That is, ethanol produced by the hydrolysis of ethene is derived from a non-renewable resource. Extraction and processing of alkanes for hydrocarbon deposits also produce greenhouse gases that are responsible for global warming.
- The production of ethanol by fermentation uses a renewable resource, glucose. Glucose is produced by photosynthesis in plants. Plants used to produce ethanol can be regrown quickly and are a renewable resource. (1 mark)

End of section 2

Question 38

- a** The equivalence point is the point at which stoichiometric amounts of reactants have been reacted*.

The end point in a titration is the point when an indicator changes colour*. (2 marks)

- b i** phenolphthalein*

The equivalence point of a titration between a weak acid (KHP) and a strong base (NaOH) occurs at a pH greater than 7*. (2 marks)

The pH range for the colour change of phenolphthalein is greater than 7.

- ii** colourless to pink (1 mark)

- c** Criterion for selection of a primary standard are:

- high level of purity
- known composition
- stable
- high molar mass
- readily soluble.

(1 mark for any one of the above)

- d** $\text{KHP(aq)} + \text{NaOH(aq)} \rightarrow \text{KPNa(aq)} + \text{H}_2\text{O(l)}$

$$\begin{aligned} n(\text{NaOH}) &= n(\text{KHP})^* \\ &= \frac{0.04814}{204.2} \\ &= 2.357 \times 10^{-4} \text{ mol}^* \end{aligned}$$

$$\begin{aligned} c(\text{NaOH}) &= \frac{n}{V} \text{ (V in litres)} \\ &= \frac{2.357 \times 10^{-4}}{0.02362} \\ &= 0.009979 \text{ mol L}^{-1}^* \end{aligned} \quad (3 \text{ marks})$$

- e** $\text{H}_2\text{SO}_4 + 2\text{NaOH(aq)} \rightarrow \text{Na}_2\text{SO}_4(\text{aq}) + 2\text{H}_2\text{O(l)}$

In the 100 mL sample of wine:

$$\begin{aligned} n(\text{SO}_2) &= n(\text{H}_2\text{SO}_4) = \frac{n(\text{NaOH})^*}{2} \\ &= \frac{0.009979 \times 0.01344}{2} \\ &= 6.705 \times 10^{-5} \text{ mol}^* \end{aligned}$$

$$\begin{aligned} m(\text{SO}_2) &= 6.705 \times 10^{-5} \times 64.07 \\ &= 0.004296 \text{ g in 100 mL sample of wine}^* \\ &= 0.004296 \times \frac{1000}{100} \\ &= 0.04296 \text{ g in 1 litre sample of wine}^* \\ &= 43.0 \text{ mg L}^{-1} \end{aligned}$$

$$\text{SO}_2 \text{ content} = 43.0 \text{ mg L}^{-1}^* \quad (5 \text{ marks})$$

Question 39

- a** In excess oxygen, all the C in the sample is converted to CO₂.

$$n(\text{C}) = n(\text{CO}_2)$$

$$= \frac{0.4538}{22.71}$$

$$= 0.01998 \text{ mol}^*$$

$$m(\text{C}) = 0.01998 \times 12.01$$

$$= 0.2400 \text{ g}^*$$

$$m(\text{H}) = 0.3597 \times \frac{2.016}{18.02}$$

$$= 0.04025 \text{ g}^*$$

$$m(\text{O}) = \text{mass sample} - m(\text{C}) - m(\text{H})^*$$

$$= 0.6000 - 0.2400 - 0.04025$$

$$= 0.3197 \text{ g}^*$$

	C	:	H	:	O
mass	0.2400	:	0.04025	:	0.3197
mole	$\frac{0.2400}{12.01}$	:	$\frac{0.04025}{1.008}$	:	$\frac{0.3197}{16.00}$
=	0.01998	:	0.03993	:	0.01998*
mole ratio	$\frac{0.01998}{0.01998}$	:	$\frac{0.03993}{0.01998}$	:	$\frac{0.01998}{0.01998}$
=	1.000	:	1.998	:	1.000*
simplest ratio	1	:	2	:	1*

empirical formula CH₂O*

(10 marks)

b $n(\text{compound}) = \frac{0.7564}{22.71}$
 $= 0.03331 \text{ mol}^*$

$$\text{molar mass, } M = \frac{m}{n}$$

$$= \frac{2.000}{0.03331}$$

$$= 60.0 \text{ g mol}^{-1*}$$

molar mass = $x \times$ mass of empirical formula unit

$$x = \frac{60.0}{30.03}$$

$$= 2 \text{ (x must be a whole number)}^*$$

Therefore, the molecular formula is C₂H₄O₂*.

(4 marks)

c

	Compound I	Compound II
Structural formula		
Name	ethanoic acid*	methyl methanoate*

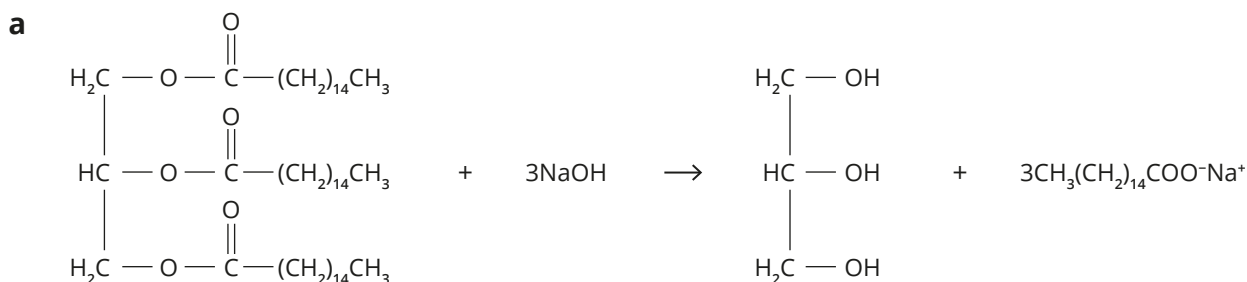
(4 marks)

Description of test	Test results for compound I (ethanoic acid)	Test results for compound II (methyl methanoate)
Test 1: Add a solution of Na_2CO_3 to each compound.	Gas bubbles will appear.	No reaction
Test 2: Test pH of a solution of each compound with universal indicator paper.	Paper indicates an acidic pH.	Paper indicates that the solution is neutral.

Other possible tests include carefully smelling each compound*. Ethanoic acid has a vinegary odour and methyl methanoate a fruity odour*.

(6 marks for valid tests, with 1 mark for the procedure and 1 mark for correctly predicting the results of the test on each compound)

Question 40

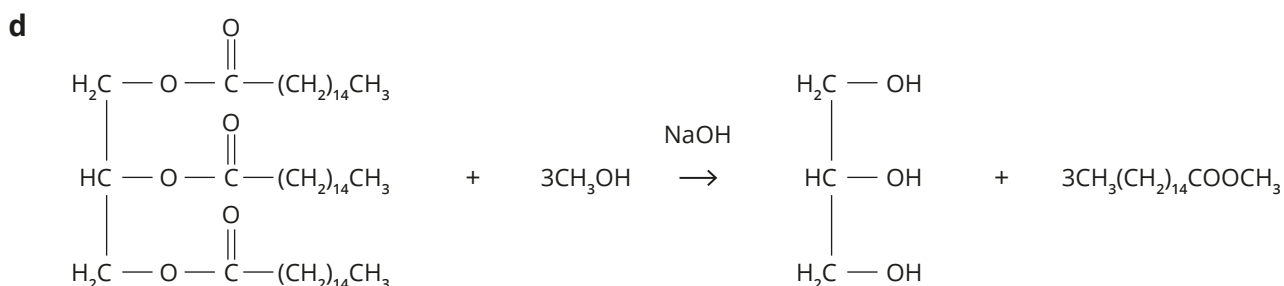
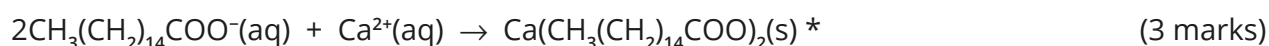


(1 mark for correct reactants and products, 1 mark for correctly balanced)

- b
- Soaps consist of the anions of long chain fatty acids*.
 - The hydrocarbon tail is hydrophobic*.
 - The hydrocarbon tail form bonds with molecules in fatty substances and does not form bonds with water molecules**.
 - The negatively charged end of the soap anion forms ion-dipole attractions with the polar water molecules*.
 - Hydrophobic ends encase the grease particles, forming a micelle*.

(6 marks)

- c
- Hard water contains Mg^{2+} , Ca^{2+} and Fe^{3+} *. These ions form an insoluble scum with soap ions but not with those of detergents*.



(2 marks)

- e
- Transesterification requires a catalyst. The base-catalysed method uses NaOH or KOH. The lipase method uses the enzyme lipase*. The lipase method does not require the high temperatures and pressures used in the base-catalysed method. It produces a higher yield and the enzyme lipase is not harmful to humans or the environment*. The bases used in the base-catalysed method are corrosive and need to be neutralised before disposal. However, since lipase is far more expensive than NaOH, the base-catalysed method is still preferred*.

(3 marks)

Question 41

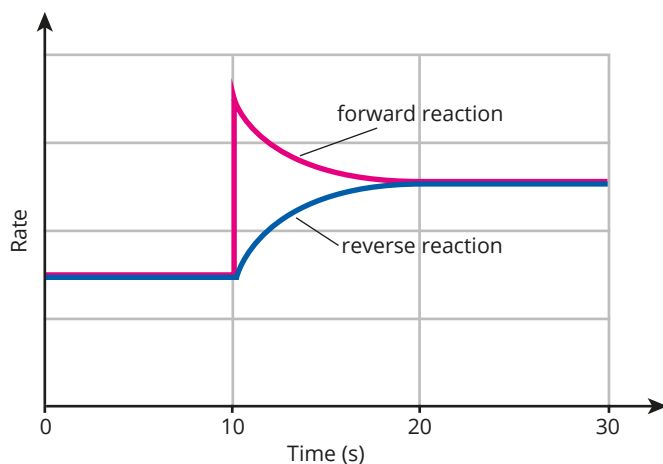
a i

Time	Explanation
between 0 and 10 seconds	The reaction system is at equilibrium. The rate of the forward reaction is constant because the equilibrium concentrations of Fe^{3+} and SCN^- are constant.
at 10 seconds	$\text{Fe}(\text{NO}_3)_2$ is added at 10 seconds. It dissolves rapidly, causing a sudden increase in the concentration of Fe^{3+} . This results in a rapid increase in the rate of the forward reaction.
between 10 and 20 seconds	The rate of the forward reaction decreases as Fe^{3+} is consumed and its concentration decreases.
between 20 and 30 seconds	Equilibrium is re-established at 20 seconds and concentration of Fe^{3+} rate remains constant, consequently the rate also remains constant.

(4 marks)

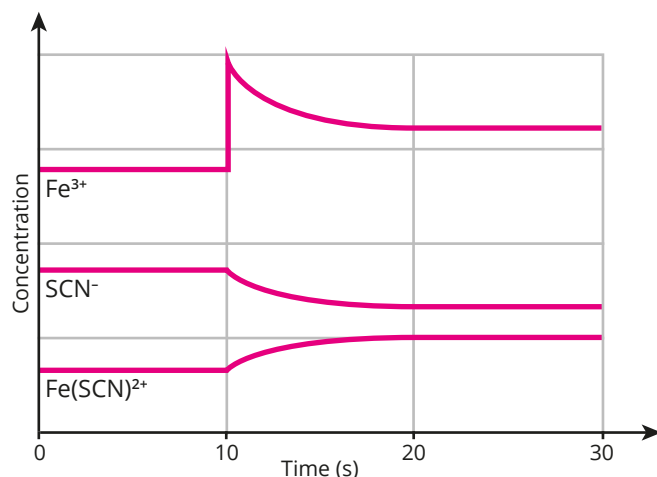
- ii The rate of the forward reaction after 20 seconds is higher than the rate between 0 and 10 seconds because the equilibrium concentration of Fe^{3+} is higher. (1 mark)

b



(3 marks)

c



(3 marks)

- d Any shift in equilibrium position can be detected by a change in the intensity of the red colour due to the FeSCN^{2+} ion. If the forward reaction is favoured, the red colour becomes more intense. If the reverse reaction is favoured, the red colour becomes less intense*.

The colour change when a solution of the equilibrium mixture is heated can be used to determine whether the reaction is endothermic or exothermic*.

If the reaction is exothermic, an increase in temperature will favour the reverse reaction*.

The concentration of $\text{Fe}(\text{SCN})^{2+}$ decreases and the intensity of the red colour is less*.

If the reaction is endothermic, the position of equilibrium favours the forward reaction and the intensity of the red colour increases*.

(5 marks)

Question 42

a i

Change	Effect on yield of SO ₂	Effect on rate of production of SO ₃
increase in temperature at constant volume	decrease	increase
decrease in pressure at constant temperature	decrease	decrease
addition of a catalyst at constant temperature and volume	no change	increase

(6 marks)

- ii Le Châtelier's principle states that if an equilibrium system is subjected to a change, the system will adjust to partially oppose the effects of the change*.

The equilibrium system will respond to a decrease in pressure by moving in a direction that will increase the pressure (i.e. the direction that favours the production of more particles, which in this case is the reverse reaction)**.

(3 marks)

- iii The production of SO₂ is an exothermic reaction, releasing heat to the surroundings. According to Châtelier's principle, the system will respond to the addition of heat by favouring the reverse endothermic reaction*.

When the system is heated, the rates of both the forward and the reverse reactions increase. However, the rate of the reverse endothermic reaction increases more than the rate of exothermic reaction. When equilibrium is re-established, the rates of the forward and reverse reactions are again equal but faster than at the lower temperature**.

(3 marks)

- b The conditions for a high yield of SO₃ are:

- high pressure
- low temperature*.

The rate of production of SO₃ is the greatest of:

- high pressure
- high temperature*.

High pressures give the best yield and rate. However, high pressure conditions are costly and it is more economical to operate the plant at lower pressure*.

A compromise temperature (400°C) produces an economically acceptable yield at an economical rate*.

(4 marks)

End of answers